

Srinivas Sivaratri

Product Engineer | Ships features, measures them, publishes the numbers
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srinivassivaratri.is-a.dev | Remote, India (IST, UTC+5:30) | Open to full-time and contract

Summary

Product engineer with an MS in Computer Science. Finds what stops a feature from being used, builds the smallest thing that fixes it, and measures the result, including when the result loses to a baseline. Upstream work reviewed by better-auth and go-ethereum maintainers.

Skills

Product: activation and cold-start loops, invite and share flows, evaluation design, baselines, honest metrics

AI: LLM agents and function calling, RAG, BM25, embeddings, Reciprocal Rank Fusion, cross-encoder reranking

Engineering: TypeScript, Next.js, React, Python, Go, PostgreSQL, Supabase, Bun

Delivery: pytest, Jest, bun test, GitHub Actions, Docker, gitleaks

Experience

Software Engineer, Independent | Remote May 2025 to present

- Traced an email-enumeration leak in better-auth to its Have I Been Pwned plugin and proposed a request-scoped skip flag with a regression test. Maintainers adopted the approach and shipped it (PR #5998).
- Designed "geth db truncate-freezer" for go-ethereum (PR #31351), revised across three core-maintainer review rounds to batch-process instead of holding about 15 GB in memory. The core team later built its own version.

Software Engineer, Accenture | Bangalore, India July 2021 to July 2022

- Wrote a Python ETL script that automated data processing, saved 50 labor hours per month, and cut errors.
- Implemented a GitHub Actions CI/CD pipeline.

Projects

Rashomon | Next.js, TypeScript, Bun, Supabase Realtime, PostgreSQL rashomon-ii0a.onrender.com | [repo](#)

- Live reading rooms where two readers discuss the same passage for five minutes. A room is useless with one person in it, so I built the paths that bring in the second reader: an invite link to the exact text, a bookmarklet, and a `/dashboard?url=` deep link that imports on arrival. Login and onboarding return you to the shared page.
- Before launch, closed the holes that made it unsafe: identity taken from a client-sent `userId`, wildcard CORS, SSRF in URL import, no row-level security. Then shipped it on free tiers (Render, Supabase, Gemini OCR in place of billed Google Vision). 41 API tests at about 98% line coverage and 32 frontend tests; CI is green after failing on every push since October 2025.

Bingo | Python, BM25, sentence-transformers, cross-encoders github.com/sivaratrisrinivas/Bingo

- Movie search over 5,000 films with keyword, semantic, hybrid (RRF) and cross-encoder-reranked retrieval, plus a side-by-side web view that shows all four rankings for one query next to their eval scores.
- On 834 known-item queries (test split), reranked hybrid put the target movie in the top 10 91.6% of the time, BM25 86.3%, plain hybrid 83.7% (below BM25 alone) and semantic 62.6%. Cut BM25 p95 latency from 3.4 s to 5 ms.

Fixer | Python, Gemini function calling, pytest github.com/sivaratrisrinivas/Fixer

- Terminal coding agent that reads the project, runs the tests, shows a diff and writes only after you approve it. Fixed a path sandbox that a sibling folder with the same prefix could bypass, and a loop that dropped every tool call after the first.
- Built an offline eval of six seeded bugs that judges each run against pristine tests. A reference fix passes 6/6, a do-nothing agent 0/6, and an agent that rewrites the tests is caught 6/6. The eval runs as a CI gate.

Education

MS, Computer Science, University of Alabama at Birmingham, AL, USA Aug 2023 to May 2025

BS, Computer Science, BV Raju Institute of Technology, India Aug 2017 to June 2021

Boot.dev, coursework, HTTP Servers in Go, RAG, CI/CD with GitHub Actions and Docker, Memory Management in C, SQL